Basic principles of trauma



Contents :

1) Definitions
2) Orthopedic terminology and fracture classification
3) Bone structure and healing
4) General Principles of fracture management
5) Complications of fractures

Fracture:

- Is a break in the continuity of bone substance.



Diaphyseal fracture

fracture of shaft of long bones



Intra- articular fracture

- fracture that involves the articular surface of a bone.



Simple fracture

The bone is broken into two fragments only.



Comminuted fracture

Bone is broken into more than two fragments



Closed fracture

fracture with intact skin or with superficial wounds.



Open fracture

fracture associated with a wound communicating the fracture hematoma with the exterior.



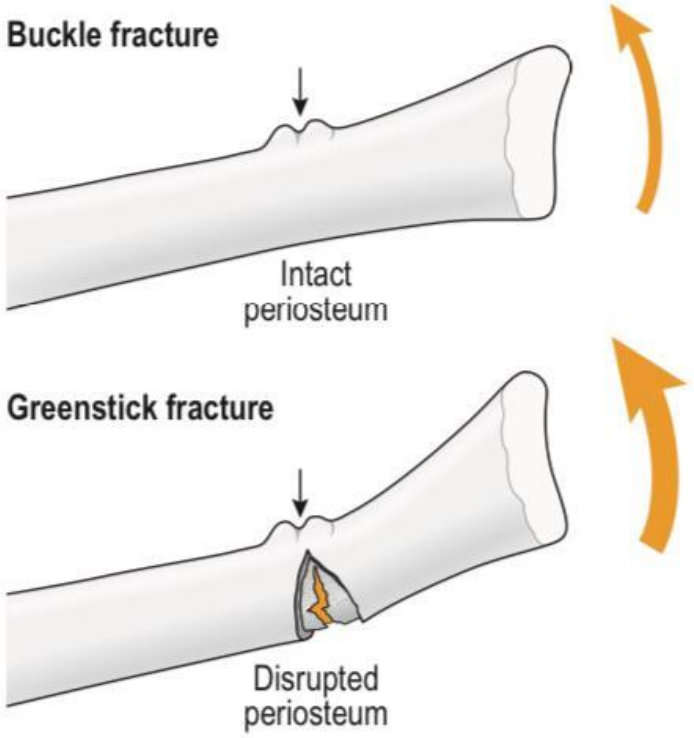
Pathological fracture

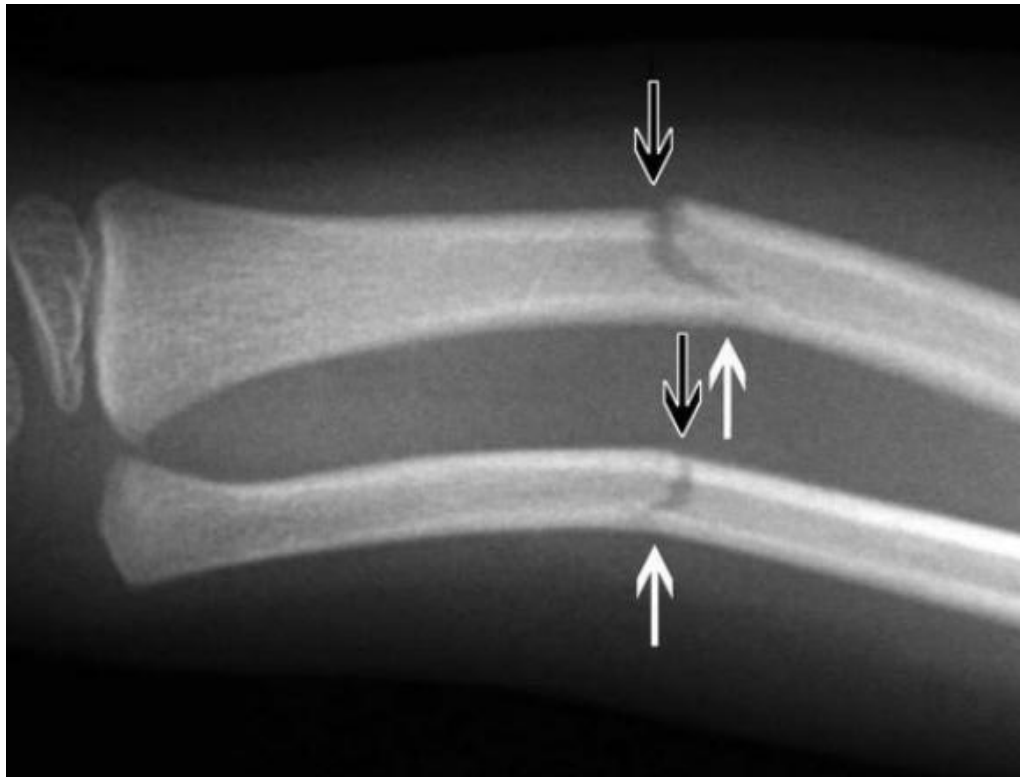
- Is a fracture occurring in a diseased or abnormal bone. It occurs after trivial trauma as the force needed to produce fracture is markedly reduced.



Greenstick Fracture

- incomplete fracture that occurs in children. One cortex breaks, the other bends and its covering periosteum remain intact.





Dislocation

The two articulating surfaces of a joint are no more in contact.

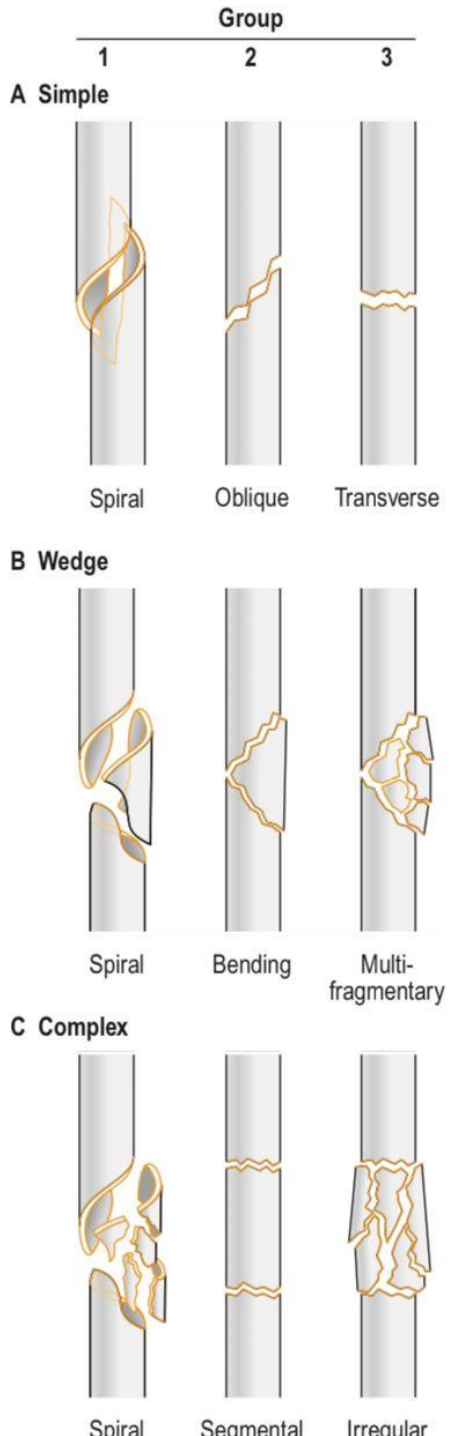


Fracture Dislocation

Is a dislocation associated with a fracture of one or more of the articulating bones.



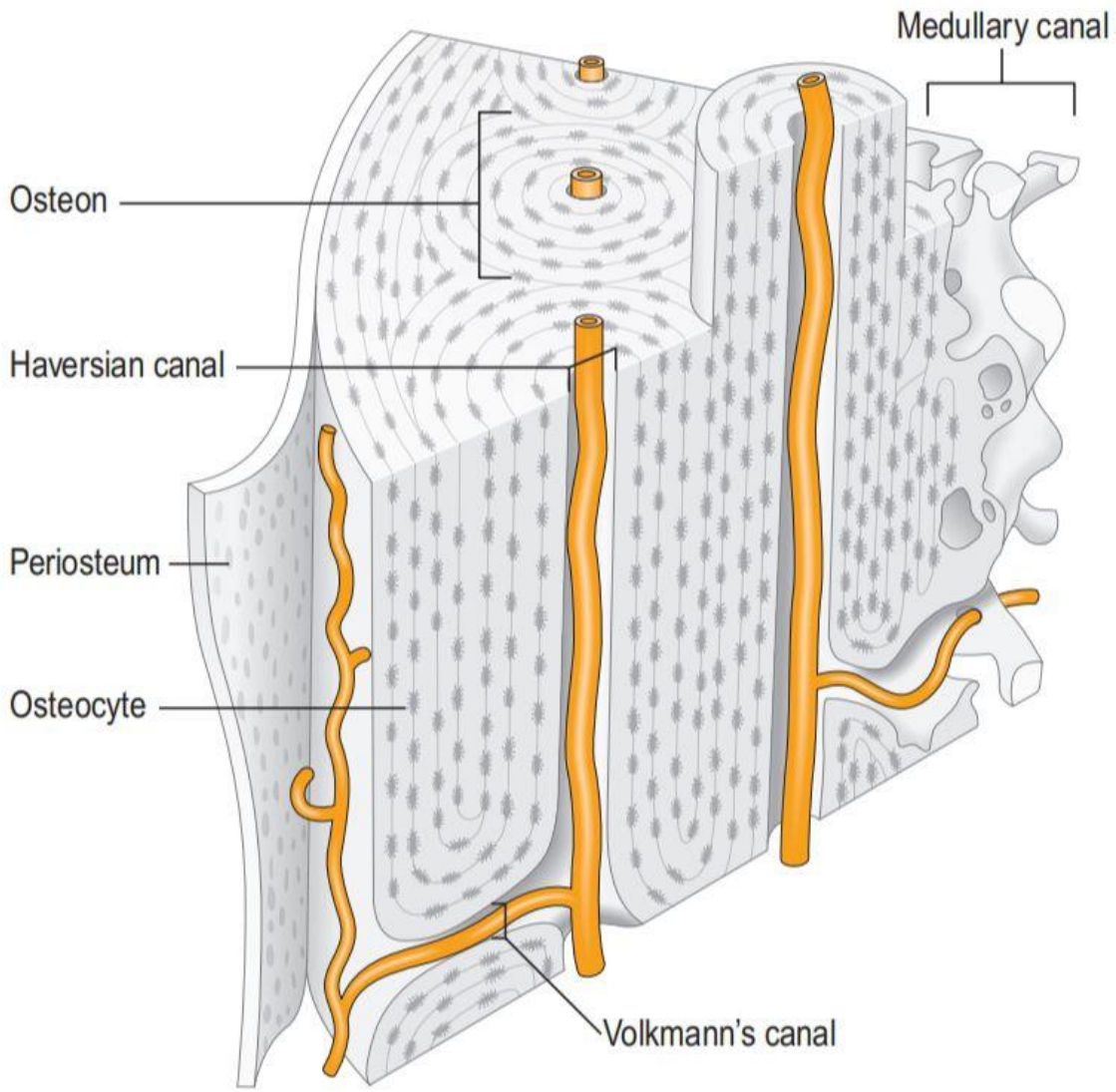
Classification of shaft fractures



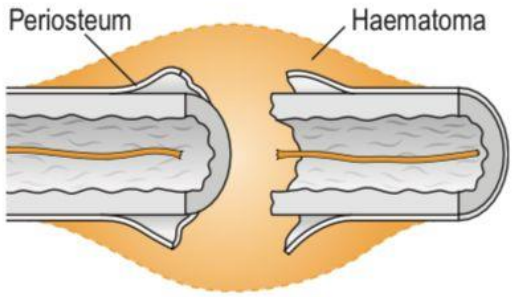
Bone anatomy

- Cortical bone is made up of components termed osteons.
- At the core of the osteon is a central Haversian canal containing blood vessels. Around the canal is a series of concentric lamellae or rings of bone matrix, embedded within which are the bone cells: osteocytes.

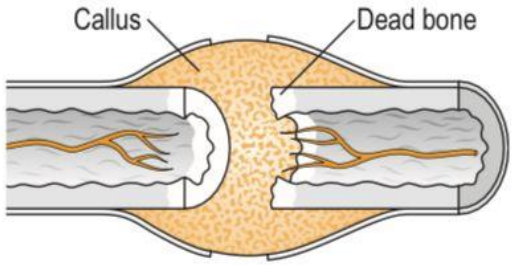
- Adjacent osteons communicate via channels termed Volkmann's canals.



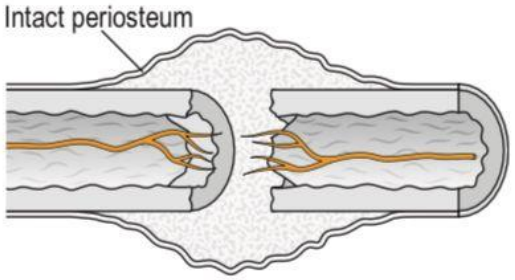
1. **Inflammation**: week 1



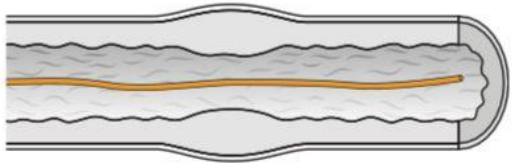
2. **Soft callus**: weeks 2–3



3. **Hard callus**: weeks 4–12



4. **Remodelling**: months to years

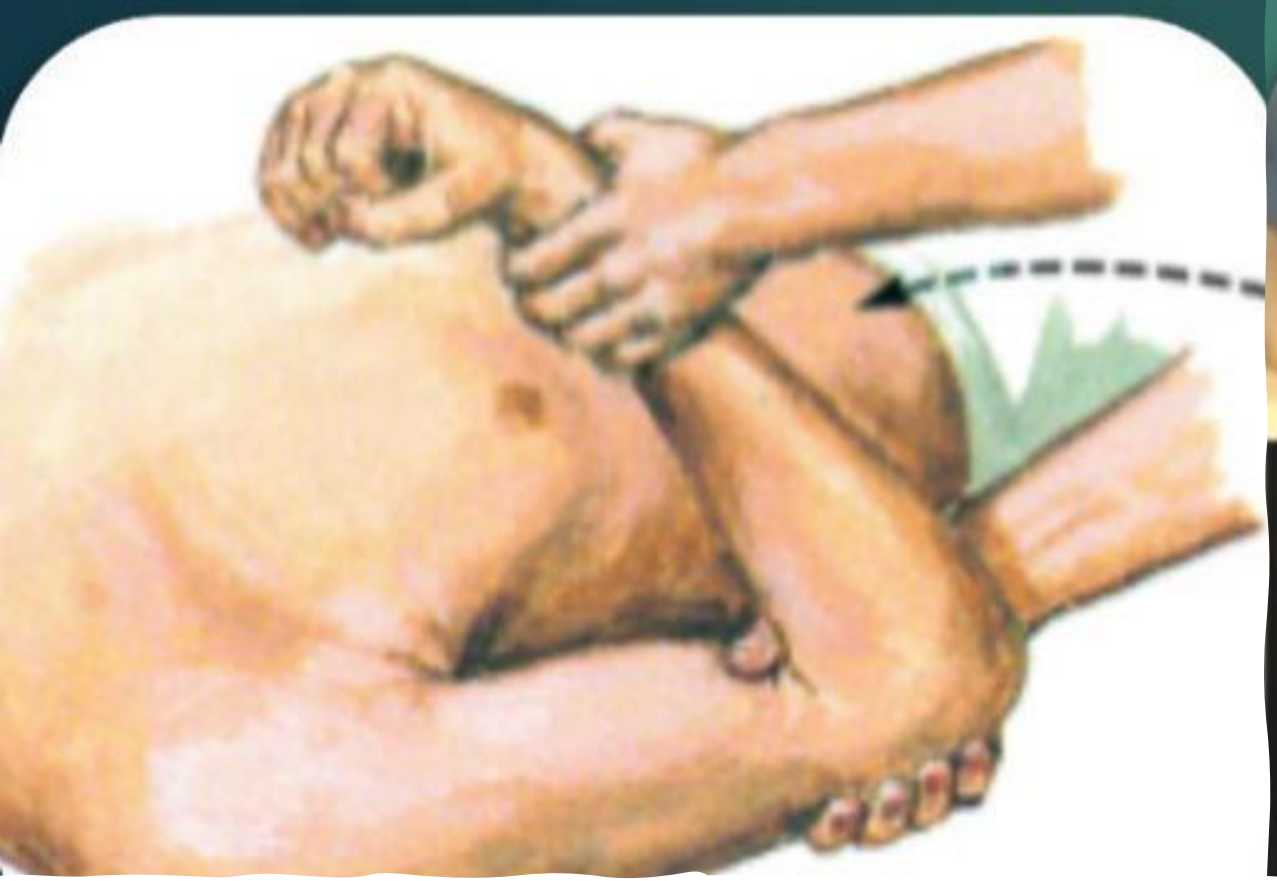


Secondary bone healing (healing with callus)

- four phases:
  - **Inflammation (week 1).**
  - **Soft callus (weeks 2–3).**
  - **Hard callus (weeks 4–12).**
  - **Remodeling (months to years).**

GENERAL PRINCIPLES OF FRACTURE MANAGEMENT







REDUCTION

- Closed (indirect)reduction
- Open (direct)reduction

Plaster Cast
(plaster of Paris)



- widely used especially for most children’s fractures
- complication

1) Tight cast
2) Pressure sores
3) Loose cast



External Fixation

- **Indications**

a. Open fractures III with extensive contaminated wounds

b. In polytrauma to stabilize fractures in minimal time

c. Open book pelvic fractures

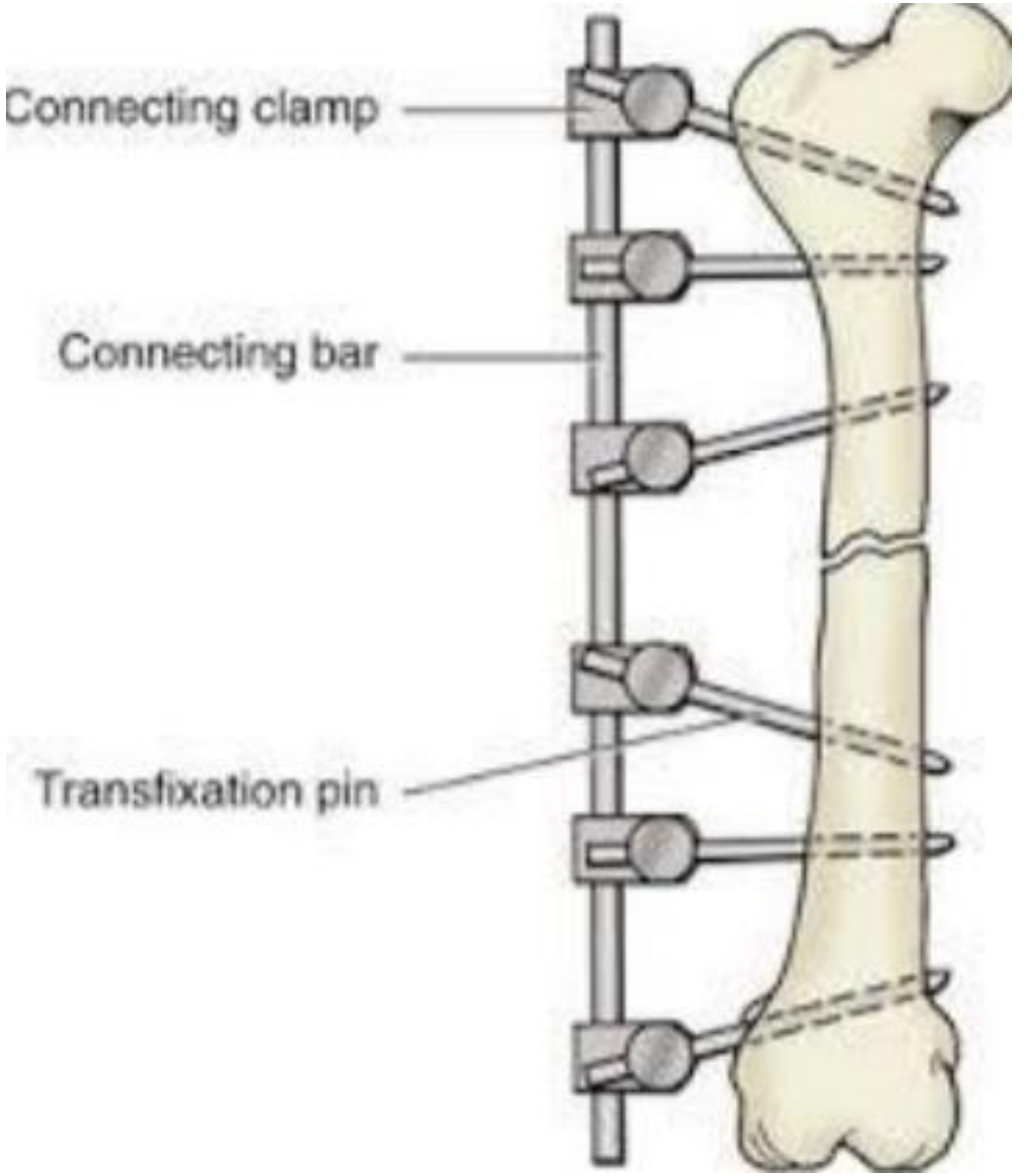
- **Disadvantages**

1) injury of nerves and vessels.

2) Pin track infection.

3) Loosening of the pins.

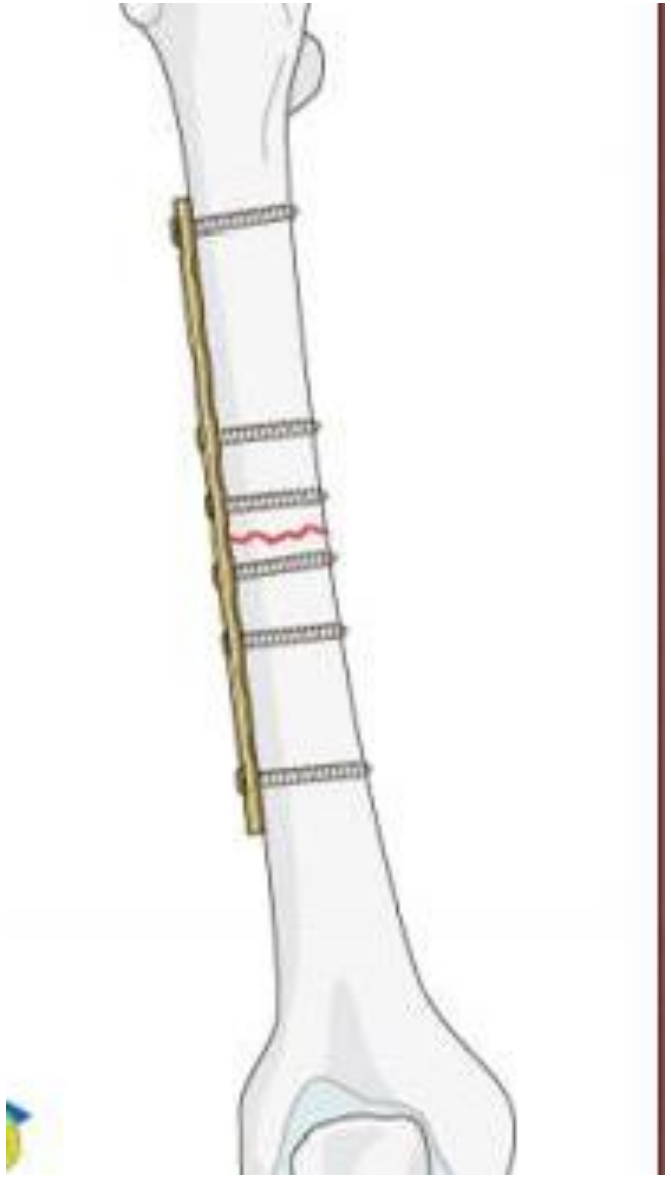
4) Transfixion of muscles resulting in stiffness of joints.



Internal Fixation

➢ **Indications:**

1. Diaphyseal fractures that cannot be closely reduced.
2. Displaced intra-articular fractures.
3. Fractures of radius and ulna in adults.
4. Displaced epiphyseal fractures.
5. Fractures in multiply injured patients to facilitate nursing.
6. Hip fractures in elderly patients to allow early mobilization of the patient.
7. Pathological fractures.
8. Fractures associated with neurovascular injuries.



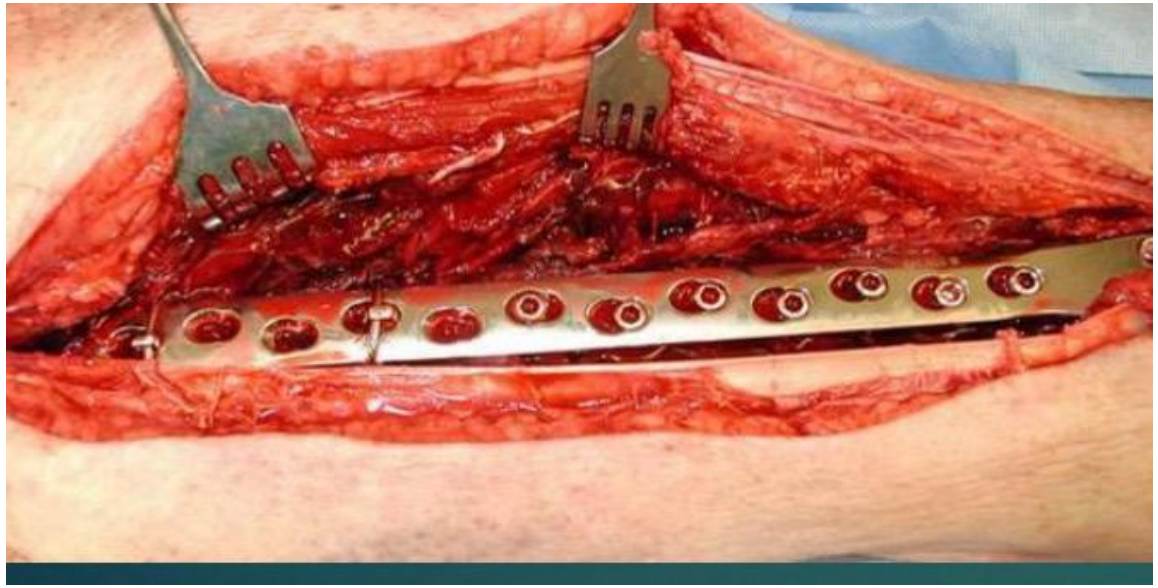
Internal Fixation

➢ **Advantages:**

- Anatomical reduction
- Stable fixation. stripping delay union
- Pain-free early joint motion
- Early functional recovery

➢ **Disadvantages:**

- Risk of infection
- Soft tissue
- Implant failure





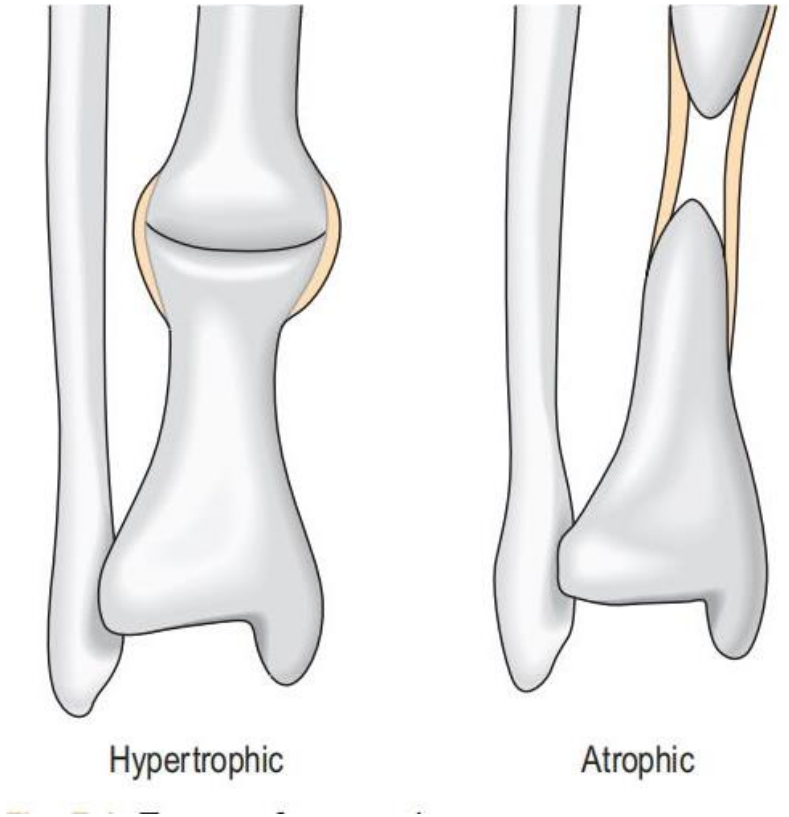


REHABILITATION

Complications of fractures

NON-UNION

a) **Hypertrophic non-union:** a mechanical problem due to inadequate fracture stability.
b) **Atrophic non-union:** a biological problem due to reduced or absent fracture healing activity, which may relate to any of the patient factors outlined above.



Complications of fractures

MALUNION

- Malunion is healing of a fracture in a non-anatomical position. This may result from inadequate initial reduction, or from subsequent loss of reduction, which may itself relate to inadequate initial stabilization of the fracture.



Complications of fractures

- **INFECTION**
- Delay fracture healing and needs :

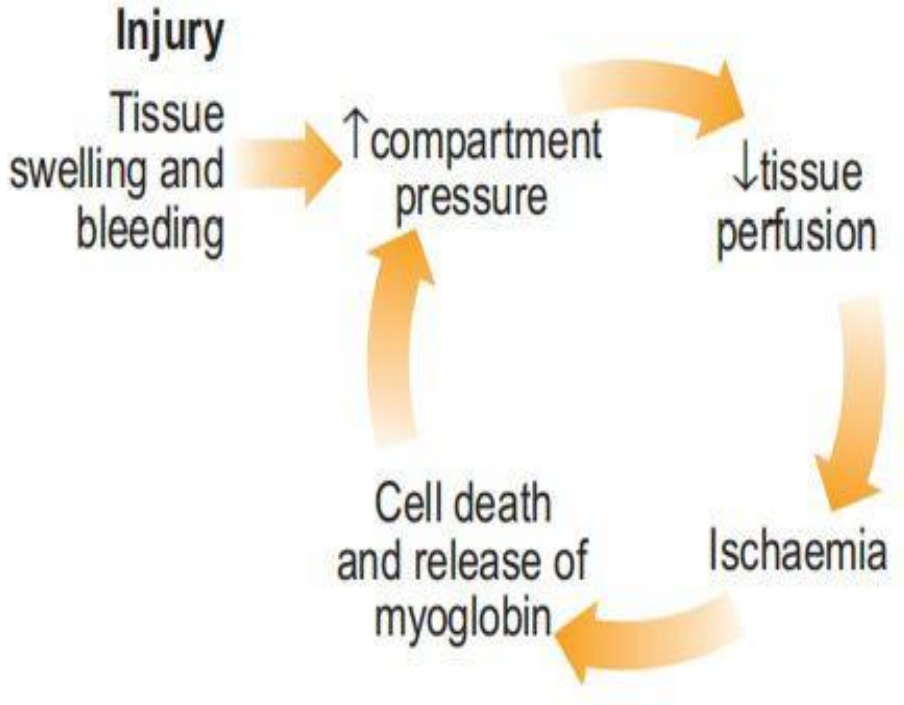
    i. **Assessment of blood indices of inflammation.**
    ii. **Wound debridement**
    iii. **Antibiotic treatment**



Complications of fractures

COMPARTMENT SYNDROME

- This catastrophic complication occurs after blunt injury to a group of muscles enclosed in a tight fascial compartment.



Complications of fractures

COMPARTMENT SYNDROME

- If this condition is suspected early, immediate splitting of the plaster cast or dressings and extension of the flexed joint are done. If pain persists, incision of the skin and deep fascia overlying all compartments in the affected part is done to relieve tension.





THANK YOU!